Definability of Non-Hermitian Homogeneous Matrix Pencils

A candidate full answer to Question 3.31 of arXiv:2210.03862

Automated mathematical discovery run `fa_banach_001`, lane 15

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Status. Candidate full solution, likely valid, requiring expert review. The bounded literature search described below found no direct later answer.

Abstract

Let p be an arbitrary homogeneous linear matrix $*$-polynomial, not assumed Hermitian. We prove that, uniformly over all operator systems E, the tuples $X \in E_k^n$ for which $1 \succeq p(X)$ form a definable set. The missing ingredient beyond the Hermitian case in Sinclair's Proposition 3.29 is a uniform projection onto the kernel of the skew-Hermitian part of the pencil. Since the pencil has a fixed finite scalar coefficient space, this projection amplifies with a norm bound independent of E. A final positive rescaling then gives the required uniform stability of the zero set.

1 Source question

Sinclair asks whether the Hermitian-pencils result immediately preceding the question extends to an arbitrary homogeneous linear matrix $*$-polynomial. The source convention $1 \succeq p(X)$ means that $1 - p(X)$ belongs to the positive cone, so the condition itself forces $p(X)$ to be Hermitian.

2 Statement

Fix the degree d, the tuple length n, and the matrix level k. For an operator system E, write $\mathcal{C}_{dk}(E)$ for the positive cone in $M_{dk}(E)$.

Theorem 1. *Let $p(x_1, \ldots, x_n)$ be any homogeneous linear matrix $*$-polynomial of degree d. The uniform assignment*

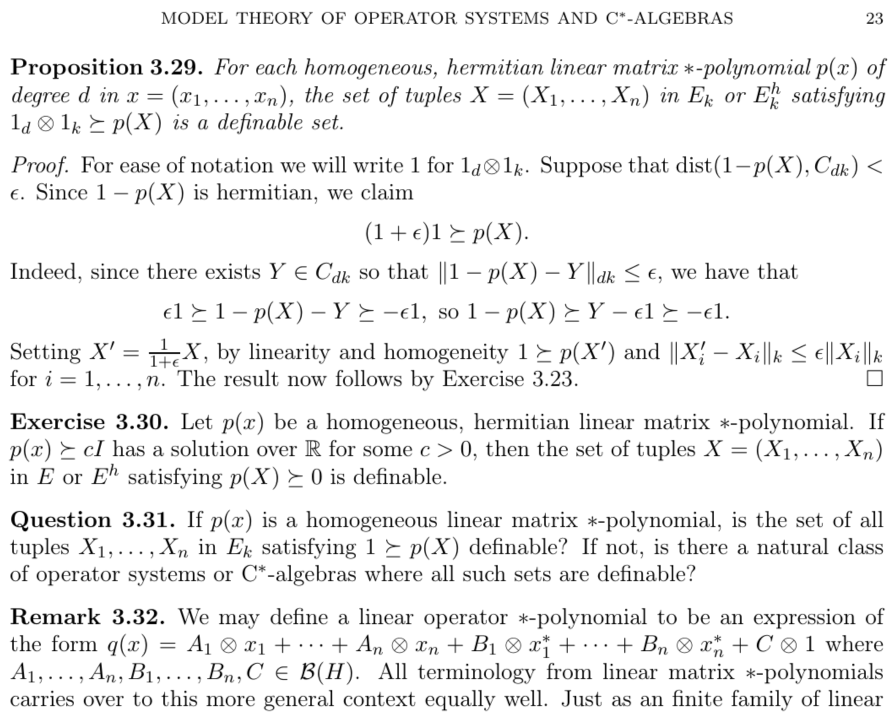
$$S_p(E) = \{X \in E_k^n : 1_d \otimes 1_k - p(X) \in \mathcal{C}_{dk}(E)\}$$

is definable over the class of all operator systems.

The assertion is understood on every fixed product of bounded sorts, exactly as in the source's treatment of unbounded uniform assignments.

3 A fixed-coefficient projection lemma

For a pencil a, denote by a_E its evaluation map on tuples over E.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Proposition 3.29. *For each homogeneous, hermitian linear matrix $*$-polynomial $p(x)$ of degree d in $x = (x_1, \dots, x_n)$, the set of tuples $X = (X_1, \dots, X_n)$ in E_k or E_k^h satisfying $1_d \otimes 1_k \succeq p(X)$ is a definable set.*

Proof. For ease of notation we will write 1 for $1_d \otimes 1_k$. Suppose that $\operatorname{dist}(1-p(X), C_{dk}) < \epsilon$. Since $1 - p(X)$ is hermitian, we claim

$$(1+\epsilon)1 \succeq p(X).$$

Indeed, since there exists $Y \in C_{dk}$ so that $\|1 - p(X) - Y\|_{dk} \le \epsilon$, we have that

$$\epsilon 1 \succeq 1 - p(X) - Y \succeq -\epsilon 1, \text{ so } 1 - p(X) \succeq Y - \epsilon 1 \succeq -\epsilon 1.$$

Setting $X' = \frac{1}{1+\epsilon} X$, by linearity and homogeneity $1 \succeq p(X')$ and $\|X_i' - X_i\|_k \le \epsilon \|X_i\|_k$ for $i = 1, \dots, n$. The result now follows by Exercise 3.23. $\square$

Exercise 3.30. Let $p(x)$ be a homogeneous, hermitian linear matrix $*$-polynomial. If $p(x) \succeq cI$ has a solution over $\mathbb{R}$ for some $c > 0$, then the set of tuples $X = (X_1, \dots, X_n)$ in E or E^h satisfying $p(X) \succeq 0$ is definable.

Question 3.31. If $p(x)$ is a homogeneous linear matrix $*$-polynomial, is the set of all tuples $X_1, \dots, X_n$ in E_k satisfying $1 \succeq p(X)$ definable? If not, is there a natural class of operator systems or C*-algebras where all such sets are definable?

Remark 3.32. We may define a linear operator $*$-polynomial to be an expression of the form $q(x) = A_1 \otimes x_1 + \dots + A_n \otimes x_n + B_1 \otimes x_1^* + \dots + B_n \otimes x_n^* + C \otimes 1$ where $A_1, \dots, A_n, B_1, \dots, B_n, C \in \mathcal{B}(H)$. All terminology from linear matrix $*$-polynomials carries over to this more general context equally well. Just as an finite family of linear

Figure 1: Proposition 3.29 and Question 3.31 in the source PDF, p. 23.

Lemma 1. *Let a be a fixed homogeneous Hermitian linear matrix $*$-polynomial. For every fixed matrix level k there are real-linear maps*

$$P_E : E_k^n \longrightarrow E_k^n$$

and a constant $C < \infty$, independent of the operator system E, such that

$$a_E(P_E X) = 0, \qquad \max_j \|(P_E X)_j - X_j\|_k \le C \|a_E(X)\|_{dk}.$$

Proof. Take real and imaginary parts of every variable. The self-adjoint part of $M_k(E)$ is, as a real vector space, the tensor product of E^h with a fixed finite-dimensional real scalar coefficient space. Thus there are finite-dimensional real spaces V, W and a fixed real-linear map $T : V \to W$ such that, under the canonical matrix-coordinate identifications,

$$a_E = T \otimes \mathrm{id}_{E^h}.$$

Choose a complement V_0 of $\ker T$ in V. Extend the inverse of $T|_{V_0} : V_0 \to \operatorname{ran} T$ to a real-linear map $S : W \to V$, and put $P = I_V - ST$. Then $TP = 0$ and

$$v - Pv = STv.$$

Amplify these fixed scalar-coordinate maps by id_{E^h}. A linear map between fixed finite-dimensional matrix coefficient spaces is a finite sum of scalar coordinate maps; hence its amplification has a norm bounded by a constant depending only on that map and on k, not on E. Therefore $P_E = P \otimes \mathrm{id}_{E^h}$ has the asserted properties, with $C = \sup_E \|S \otimes \mathrm{id}_{E^h}\| < \infty$. $\square$

Why this lemma is needed. Writing a condition as the intersection of a Hermitian inequality and a linear equality identifies the correct set, but zero sets of definable predicates are not automatically definable in continuous logic. Lemma 1 supplies the missing uniform perturbation estimate.

4 Proof of Theorem 1

Define the Hermitian homogeneous pencils

$$q = \frac{p + p^*}{2}, \qquad r = \frac{p - p^*}{2i},$$

so that $p = q + ir$. Consider the definable predicate

$$f_E(X) = \operatorname{dist}\big(1 - p(X), \mathcal{C}_{dk}(E)\big).$$

Its zero set is exactly $S_p(E)$. By the uniform zero-set criterion stated as Exercise 3.23 in the source, it suffices to show on each bounded product of sorts that $f_E(X) \to 0$ forces the distance from X to $S_p(E)$ to tend to zero, with a modulus independent of E.

Suppose $f_E(X) < \delta$. Up to replacing δ by an arbitrarily slightly larger number, choose $A \in \mathcal{C}_{dk}(E)$ with

$$e := 1 - p(X) - A, \qquad \|e\| < \delta.$$

Taking skew-Hermitian and Hermitian parts gives

$$\|r(X)\| < \delta, \qquad \|1 - q(X) - A\| < \delta.$$

In particular,

$$q(X) \preceq (1 + \delta)1.$$

Apply Lemma 1 to r and set $Y = P_E X$. Then

$$r(Y) = 0, \qquad \max_j \|Y_j - X_j\| \le C\delta.$$

Let L be a fixed Lipschitz constant for the evaluation map q_E, uniform in E; such a constant is obtained by summing the norms of the finitely many scalar coefficient matrices. It follows that

$$q(Y) \preceq \big(1 + (1 + LC)\delta\big)1.$$

Put $K = 1 + LC$ and

$$Z = \frac{Y}{1 + K\delta}.$$

Homogeneity yields $r(Z) = 0$ and $q(Z) \preceq 1$. Hence $p(Z) = q(Z)$ and $1 - p(Z) \in \mathcal{C}_{dk}(E)$, so $Z \in S_p(E)$.

On a product of sorts with $\max_j \|X_j\| \le R$,

$$\begin{aligned}\max_j \|Z_j - X_j\| &\le C\delta + \frac{K\delta}{1 + K\delta} \max_j \|Y_j\| \\ &\le C\delta + K\delta(R + C\delta),\end{aligned}$$

which tends to zero uniformly in E. The zero-set criterion proves that S_p is definable. This answers both parts of Question 3.31: no restriction to a smaller natural class is necessary. □

5 Proof intuition and failure modes

Near-positivity of $1 - p(X)$ already controls two errors. Its Hermitian part says that the order inequality is nearly true, while its skew-Hermitian part says that $p(X)$ is nearly self-adjoint. Because the skew constraint is a fixed finite system of real-linear scalar equations, "nearly in its kernel" can be repaired uniformly. Once repaired, the exact argument from the Hermitian case applies.

Two shortcuts are unsafe. First, merely intersecting the Hermitian feasible set with $\{r = 0\}$ does not establish definability. Second, rescaling before eliminating $r(X)$ never makes a nonzero skew part exactly vanish. The fixed coefficient projection resolves both issues.

6 Verification and novelty bounds

The accompanying script constructs random non-Hermitian pencils over finite matrix algebras, forms the real-linear skew map, orthogonally projects onto its kernel, and checks the final spectral rescaling. This is a sanity check of the algebraic mechanism, not a proof of the uniform amplification lemma.

A bounded search on 11 August 2026 covered the run's registry, solution, attempt, and active-claim indexes; the exact arXiv id and question phrase; the current author-hosted PDF; and searches combining non-Hermitian/Hermitian decomposition with operator-system definability. The current PDF retains the question, and no direct answer was found. This is provisional novelty evidence only.

Reference

T. Sinclair, *Model Theory of Operator Systems and C^*-Algebras*, arXiv:2210.03862 (2022); source PDF, Question 3.31 and Proposition 3.29.